

ANIRUD MOHAN

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SUMMARY

AI Engineer and MS in Applied Machine Learning with hands-on experience shipping production LLM systems end-to-end. Specialized in inference optimization vLLM serving, KV caching, quantization, and dynamic batching and in building agentic and RAG applications with LangChain/LangGraph. Comfortable owning systems across the stack, from GPU model serving to FastAPI/React delivery, and turning research techniques into reliable, measurable services.

EDUCATION

University of Maryland | College Park, MD

Aug 2024 - May 2026

Master of Science in Applied Machine Learning | GPA: 3.83/4.0

Coursework: Deep Learning, NLP, Model Optimization, Algorithms for ML, MLOps, Safety and Trustworthy AI.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

ML / AI: PyTorch, Triton, CUDA, HuggingFace Transformers, LoRA, QLoRA, Diffusers, Scikit-learn, FSDP, Deepspeed

LLM Systems: RAG Pipelines, LangChain, LangGraph, LlamaIndex, vLLM, MCP, Tensor-RT

Infrastructure: FastAPI, Docker, Kubernetes, Async Python, CI/CD, Git

Cloud & Data: GCP, Databricks, Snowflake, PostgreSQL, MongoDB

Specialized: Inference Optimization, KV Caching, Dynamic Batching, Quantization, Flash Attention, GPU Workloads

WORK EXPERIENCE

CarinaAI

Herndon, USA

AI Engineer Intern

Oct 2025 – Jan 2026

- Engineered the **AutoRegistry** feature for the DeIdentifier platform by building a production **RAG pipeline** for automated clinical registry extraction, improving accuracy from **69% to 82%**.
- Conducted **embedding and model benchmarking experiments**, evaluating retrieval performance across multiple embedding models and comparing reasoning vs non-reasoning LLMs for clinical document analysis.
- Optimized deployment using **vLLM with KV caching**, enabling efficient inference and successfully serving a **20B parameter model on a 16GB VRAM GPU** through memory-efficient configuration.

Thapovan Info Systems

Chennai, IND

Junior Machine Learning Engineer

Oct 2023 – Jun 2024

- Developed an **advanced RAG pipeline** for medical insurance knowledge bases using a **router-based workflow** to dynamically select retrieval strategies for different query types.
- Implemented structured **input/output guardrails and validation pipelines**, reducing irrelevant model responses by **40%** and improving reliability of production AI services.
- Experimented with **PEFT fine-tuning (QLoRA)** using the **MLX framework**, evaluating trade-offs between domain adaptation via fine-tuning and retrieval-based architectures.

Azentio Software

Chennai, IND

Software Developer Intern

Feb 2023 – Sept 2023

- Refactored a monolithic application into **microservices architecture**, improving scalability and modularity of backend services.
- Built Python-based **data extraction pipelines** using Scrapy and BeautifulSoup for automated ingestion and transformation of large datasets.

RELEVANT RESEARCH PROJECTS

LLM Inference Engine from Scratch | [GitHub](#)

2026

- Architected a production LLM inference server for **Qwen2.5 models** implementing **KV caching**, **INT4 NF4 quantization**, and a custom async **dynamic batching engine**, saturating **multiple GPUs (>90% utilization)** while serving **13,558 requests/hour**.
- Benchmarked core inference optimizations in PyTorch including KV caching (**10× speedup**), Sliding Window Attention (**16× memory reduction**), Flash Attention, and Naive Paged Attention, documenting hardware-specific trade-offs.
- Conducted architecture experiments rejecting speculative decoding in favor of **dynamic batching**, achieving **3.7× latency reduction (100s to 27s)** while eliminating inference failures.
- Extending the system with a **custom continuous batching scheduler** inspired by vLLM to dynamically merge concurrent token generation workloads and further improve GPU utilization.

FinQA: Financial Q&A Reasoning Agent | [GitHub](#)

2026

- Built a financial-document Q&A **agent over S&P 500 earnings reports** using a **LangGraph** multi-step pipeline (retrieve → plan → generate executable DSL → answer), achieving **65% execution accuracy** and **56% program accuracy** on a 200-sample benchmark.
- Served **Qwen3-8B (FP8)** via vLLM behind a FastAPI backend with a React frontend, grounding numerical reasoning in retrieved table/text chunks through a **ChromaDB** vector store.
- Hardened reliability with a **33-test Pytest evaluation harness** covering retrieval, DSL execution, and end-to-end accuracy.

Deep Research Agent | [GitHub](#)

2026

- Engineered a multi-source research **agent from scratch**, decomposing queries, searching the web via **Tavily**, and synthesizing answers under a **hard 2,048-token-per-call memory budget** enforced with **tiktoken**.
- Designed a **3-tier hybrid memory architecture** (ChromaDB vector store + 3-level summarization cascade + FIFO working buffer) enabling precise follow-up retrieval while staying within budget.
- Implemented a cost-aware **two-model strategy** (gpt-5-mini for reasoning-heavy synthesis, gpt-4o-mini for utility calls), holding each research session **under \$0.05** across ~13 LLM calls.